

SOF IMO — Topic Deep-Dive: Measurement

For Ambar, Class III CBSE

1. The three families

We measure	Small unit	Big unit	Conversion
Length / distance	centimetre (cm), metre (m)	kilometre (km)	1 m = 100 cm; 1 km = 1000 m
Weight	gram (g)	kilogram (kg)	1 kg = 1000 g
Capacity (liquids)	millilitre (mL)	litre (L)	1 L = 1000 mL

Memory hook: **kilo = 1000** (kilometre, kilogram). The odd one out is **1 m = 100 cm** (only 100!). This is the most-tested conversion trap in Class III.

2. Choosing a sensible unit

Thing	Sensible unit
Length of a pencil, eraser, handspan	cm
Length of a saree, height of a person, classroom	m
Delhi to Agra, distance between cities	km
Weight of an eraser, a biscuit, a letter	g
Weight of a sack of rice, a watermelon, a person	kg
Medicine in a spoon, a small syrup dose	mL
Milk, water in a bottle or bucket	L

Tools: **ruler/measuring tape** for length, **weighing balance** for weight, **measuring jug** for capacity.

3. Converting

Big unit → small unit: MULTIPLY.

- $5 \text{ m} = 5 \times 100 = 500 \text{ cm}$
- $4 \text{ km} = 4 \times 1000 = 4000 \text{ m}$
- $3 \text{ kg} = 3000 \text{ g}$; $2 \text{ L} = 2000 \text{ mL}$

Mixed units: convert the big part, then ADD the small part.

Worked example: $7 \text{ m } 25 \text{ cm} = 700 \text{ cm} + 25 \text{ cm} = \mathbf{725 \text{ cm}}$ (NOT 7025 — that mistake comes from treating m like km).

Halves: half of 1 kg = 500 g; half a litre = 500 mL; half a metre = 50 cm.

4. Adding and subtracting with units

Line up the same units, carry/borrow in thousands (or hundreds for cm).

Worked example 1 (add): $45 \text{ cm} + 38 \text{ cm} = 83 \text{ cm}$. **Worked example 2 (subtract with mixed units):** $6 \text{ kg} - 2 \text{ kg } 500 \text{ g}$. $6 \text{ kg} = 6000 \text{ g}$. $6000 - 2500 = 3500 \text{ g} = \mathbf{3 \text{ kg } 500 \text{ g}}$. **Worked example 3 (capacity):** $5 \text{ L} - 2 \text{ L } 250 \text{ mL} = 5000 - 2250 = 2750 \text{ mL} = \mathbf{2 \text{ L } 750 \text{ mL}}$.

Comparing

Convert to the SAME unit first, then compare. - 1 kg vs $750 \text{ g} \rightarrow 1000 \text{ g}$ vs $750 \text{ g} \rightarrow 1 \text{ kg}$ is heavier. - 2500 g vs $2 \text{ kg} \rightarrow 2500$ vs $2000 \rightarrow 2500 \text{ g}$ is more. - 90 cm vs $1 \text{ m} \rightarrow 90$ vs $100 \rightarrow 90 \text{ cm}$ is shorter.

5. Common traps

Trap	Wrong	Right
1 m = 1000 cm	1000	1 m = 100 cm
7 m 25 cm = 7025 cm	7025	725 cm
Adding when asked "how much left"	1 L drink 400 mL → 1400 mL	600 mL left
Comparing without converting	750 g > 1 kg "because 750 > 1"	convert: 1000 g > 750 g
Wrong unit match	weight in litres, milk in metres	weight-kg, capacity-L, length-m
100 cm vs 1 m confusion	"100 cm is longer than 1 m"	they are EQUAL

6. High-frequency IMO question types

1. Direct conversion: "1 km = ____ m", "3 kg = ____ g".
2. Best unit: "Which unit for the length of a pencil?"
3. Heavier/longer/more comparisons with mixed units (1 kg vs 750 g).
4. Add/subtract word problems: ribbons joined, sugar sold, milk left.
5. Equal cutting/filling: "20 m rope into 5 m pieces", "2 L jug with 500 mL glasses".
6. Ordering lightest to heaviest.
7. Achiever's: daily use × days, then subtract from the total (oil can problem).

7. Key facts checklist

- ☐ 1 m = 100 cm; 1 km = 1000 m
- ☐ 1 kg = 1000 g; 1 L = 1000 mL
- ☐ Half kg = 500 g; half L = 500 mL; half m = 50 cm
- ☐ Big → small unit: multiply
- ☐ Compare only after converting to the same unit
- ☐ cm for small lengths, m for medium, km for distances
- ☐ g for light things, kg for heavy; mL for spoonfuls, L for bottles/buckets

8. Quick drill (10 questions)

1. 1 m = ____ cm
2. 2 km = ____ m
3. 4 kg = ____ g
4. Half a litre = ____ mL
5. Which is heavier: 2 kg or 1800 g?
6. 3 m 40 cm = ____ cm
7. A bottle has 1 L. Ambar drinks 250 mL. How much is left?
8. A 24 m rope is cut into 6 m pieces. How many pieces?
9. Which unit: distance from Delhi to Jaipur?
10. 5 kg – 1 kg 500 g = ?

Answers: 1) 100 2) 2000 3) 4000 4) 500 5) 2 kg (2000 g) 6) 340 7) 750 mL 8) 4 9) km 10) 3 kg 500 g